Technology Stack Specification

Template: Define Your Technology Stack	Maximum Marks: 2 Marks
Date: 15 March 2026	**Team ID:** FV-AI-2026-X1
Project Name: FoodVerse AI	**Track:** AI/ML & Generative AI

Technology Stack Details

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

S.No	Architecture Layer	Technology Chosen	Justification / Purpose for FoodVerse AI
1	Frontend / Client-Side	React 18 + Vite 6 + Modern CSS Variables	Provides ultra-fast client-side hot reloading, reusable component architecture, canvas-based synthetic test image synthesis, and responsive audio recording.
2	Backend / Server-Side	Python 3.10+ & FastAPI + Uvicorn	Asynchronous ASGI framework delivering high throughput, native Pydantic v2 data validation, automated OpenAPI Swagger documentation, and non-blocking I/O.
3	Computer Vision Core	OpenCV (opencv-python-headless), NumPy, PIL	Executes noise reduction, HSV color-space discoloration masking, and Laplacian gradient texture variance calculation to detect early biological decay.
4	AI & Reasoning Engine	Groq Cloud AI (Qwen 3.8B Multimodal & Whisper Large v3 Turbo)	Delivers near-instantaneous vision-language reasoning, multi-item plate decomposition, accurate macro calorie calculation, and fast speech transcription.
5	Cloud Hosting & Deployment	Vercel (Frontend CDN) & Render (Backend Web Service)	Zero-cost, enterprise-grade cloud hosting providing continuous git-triggered deployments, SSL certificates, global edge caching, and scalable HTTP handling.
6	Version Control & Quality	GitHub + Automated Edge-Case Test Suite (FastAPI TestClient)	Maintains strict code integrity, git commit auditing, automated regression testing verifying 9/9 edge cases (corrupted images, dim lighting, extreme diets).